

1. Endure Industries Emesis Basin

The Endure Industries Emesis Basin is a kidney shaped tray that is used in surgical and clinical spaces. It is made of polypropylene (PP) with a glossy finish for easy cleaning. Each basin has a capacity of 500 milliliters and includes graduated measurement markings and lipped edges for easy handling. Prices vary depending on bulk orders. On the website, a 50 pack of trays costs 16 dollars.

[Link to source](#)



Summary:

This product is commonly used in clinics and hospitals to collect fluids, medical waste, and used dressings during surgeries or wound care. The kidney shape of the basin allows it to be placed right against the patient's body, which makes it easier to collect fluids. The sturdy polypropylene material ensures durability, while also making it easier for cleaning. The graduated measurements assist in fluid calculation.

Helpful Takeaways:

- Clear measurement markings make it easier to see and keep track of how much fluid has been collected
- Having lipped edges around the basin can make it easier to handle during medical procedures and enhance portability
- The durable material (PP) with the smooth finish can make it easy for cleaning, have a clean, sleek design, and long lasting.

Unhelpful Takeaways:

- The kidney standard shape makes it harder to handle comfortably, especially when cleaning or rinsing arms and legs during surgeries.
 - It doesn't have any of the helpful design features that were requested by Stakeholders for placing limbs or built in suction for easier fluid removal
 - Since it only comes in one size, it is not flexible enough to meet different surgical needs.
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2. Medegen Conventional “Kidney” / emesis basin (Stainless steel/molded plastic)



Summary

The classic emesis basin (kidney dish) is a concave, kidney-shaped bowl used in hospitals and clinics for vomit, wound dressings, irrigation waste, and small instrument holding. Typical materials are 304 stainless steel (reusable) or medical-grade polypropylene (PP) / polyethylene (disposable or reusable) for injection-molded basins. Typical product dimensions range from ~200–220 mm (8–8.5 in) length with capacities from ~250–800 mL depending on model. Product pages list sterilization temperatures and measurement graduations for some models.

Unique Features & Functions

- **Kidney concave geometry:** A smooth continuous curve with one side shallower to allow closer approach to a patient’s body; helps to catch fluid while allowing personnel to hand it off.
- **Materials:** Stainless steel for durability and high-temperature sterilization; molded plastics for disposability and cost. Product literature calls out steam sterilization limits and chemical-resistance.
- **Manufacturing:** Stamped and formed (metal) or injection molded (plastic); many plastic basins include subtle measurement graduations or lipped edges for carrying.

Incorporating in Design

- **Keep the kidney baseline curvature** because clinicians are already trained with that shape and it naturally shelters fluid. Use the same overall length footprint (~200–220 mm) so it fits existing OR trays and storage.
- **Smooth, radiused internal surfaces:** maintain large fillet radii ($R \geq 8\text{--}12\text{ mm}$) on interior surfaces to remove sharp contact points and reduce localized pressure against skin.
- **Material options:** offer two variants — a rigid, autoclavable stainless version for repeated OR use and a disposable molded-PP version with a removable soft liner (see later designs). Manufacturer product specs are a good baseline for dimensions and allowable sterilization regimes.

Cons

- **Shallow slope & single low point:** standard basins often have only a single low spot and shallow slope, suboptimal when a limb rests nearly horizontal because fluid pools rather than exits.
- **Pressure concentration:** the rim and unsupported interior can create localized pressure points on soft tissue when a patient's limb rests directly on the basin.
- **No integrated drainage:** most conventional kidney dishes lack integrated drain ports or suspension features, the basin must be tilted or emptied manually.

[Source](#)

[Source](#)

3. Emesis/ irrigation basin with cutouts and a side spigot (published patent)

Summary

- There are patent disclosures and design patents that show emesis or irrigation basins with U-shaped cutouts in the rim so the limb can enter the bowl cavity and integrated side spigots / drain ports to allow controlled drainage without tilting the entire basin. These are design and utility patent examples filed by medical-supply companies to improve limb access and avoid lifting the limb to empty fluid. One such published design disclosure shows multiple views and claims regarding the basin shape and drainage features.

Unique Features & Functions

- **Rim cutout / cradle:** sections removed from the rim create a lowered entry area where an arm or leg can rest more naturally into the bowl instead of sitting on top of a full rim.
- **Resting ledge:** a broad, contoured ledge within the cutout distributes weight across a larger surface area of the limb.
- **Side spigot/drain:** a threaded spigot or valve integrated into the peripheral wall that can be opened to drain fluid into a container or tubing without lifting the spigot usually requires a gasket and sealing design to avoid leaks and contamination.

Incorporating Into Design

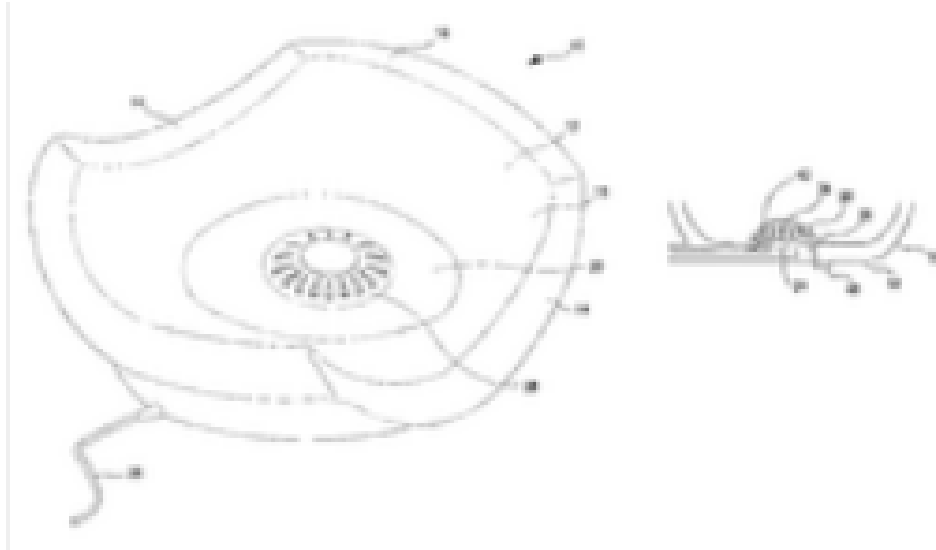
- **Incorporate a cradle/cutout sized to common limb diameters:** design multiple cradle radii small (40–60 mm) for forearms, medium (80–120 mm) for lower legs, or use a continuous, tapered cradle geometry to accept a range of limb sizes. Prototype with 3D-printed molds to test fit and comfort. Use anthropometric data for forearm / lower-leg circumferences when choosing radii.
- **Wide ledge contact** ($\geq 20\text{--}30$ mm ledge width) to distribute pressure and reduce bruising, round the ledge with a large fillet radius.
- **Add a side drain port with a removable valve** so fluid can be removed through tubing to a waste container; design the port with an internal trap geometry (small pop-out strainer) so tissue/debris won't immediately clog the port.

Cons

- **Splash & spill risk:** partial rim removal means fluid near the rim may more easily escape if the basin is bumped or the limb moves. Engineering countermeasures (sloped internal walls, lip geometry) are required.
- **Sealing & sterilization:** integrated spigot/valve adds parts that must be reliably sterilizable or be permanently disposable. Complex sealing surfaces increase manufacturing and cleaning cost.
- **Structural integrity:** removing rim material reduces stiffness, the basin must be sufficiently thick or reinforced to avoid flexing when handled.

[Source](#)

4. Extremity Debridement/Irrigation Basin With Contoured Seats + Filter (Patent)



Summary

Debridement and irrigation extremity basin is a utility patent for a basin specifically intended for extremity (limb) wound debridement and irrigation. The patent emphasizes contoured seats on opposing sidewalls that cradle a limb, plus a drainage hole plus filtering device to keep tissue and particulates from clogging the drain tubing. The patent text and drawings describe stable geometry, placement of the drain at the basin's lowest point, and removable filter elements.

Unique Features & Functions

- **Contoured opposing seats:** both side walls have shaped recesses that together stabilize and center the limb, spreading load and preventing lateral sliding. This helps keep the wound and limb in a predictable orientation relative to the drain.
- **Drain/filter assembly:** a drain port placed at or near the geometric low point with a removable screen or filter housing to stop large debris from entering tubing. This enables continuous irrigation flow or connection to suction without clogging.
- **Stable footprint:** the basin's outer geometry is designed to be broad and low to reduce tipping risk when partially filled.

Incorporating Into Design

- **Adopt dual contoured seats:** add symmetric or asymmetric cradles on each side of the basin interior sized to match average limb cross sections. Use curved contact surfaces with compliant material (foam or soft polymer) for comfort . Prototype seat curvature using 3D scans of

forearm/lower-leg cross sections or standard anthropometric tables.

- **Integrate a removable filter cartridge:** design a recessed filter basket (user-removable) sized so that the drain channel remains accessible but the basket captures particulates down to a chosen size (500–1000 μm depending on debris expected). Provide a simple twist-lock or snap-in interface for quick removal and cleaning/replacement.
- **Place drain at true low point:** design the interior slope so that even when a limb is resting in the cradle the fluid path still leads to the drain (use CFD/prototype gravity tests to verify flow). Consider a curved channel under the cradle that guides fluid around the limb to the drain.

Cons

- **Filter maintenance & clogging:** filters must be removed and cleaned/replaced, if not maintained, flow is blocked. The user workflow needs clear instructions and easy access.
- **Fit variability:** fixed contoured seats work best when limb sizes and positions fall within design tolerances; very small or very large limbs may not seat well. Consider multiple basin sizes or an adjustable/compliant seat.
- **Manufacturing complexity & cost:** more complex geometry and added filter hardware raise part cost and may complicate sterilization.

[Source](#)

5. Basin stands and adjustable mounting hardware (stands, clamps, trolley support)



Summary

- Surgical basins are often used together with solution stands, basin stands, or rail-mounted holders to position basins at a convenient height and orientation. Leading suppliers (Medline, Cardinal Health, McKesson) sell basin stands, floor stands with casters and adjustable posts or fixtures that hold a basin pan, enabling height and angle adjustment and easier waste handling. These stands are used for irrigation solution containers, waste collection, and to hold basins in position near a patient.

Unique Features & Functions

- **Height adjustability:** telescoping posts or clamp joints allow clinicians to set basin height to align with the limb or surgeon's working position.
- **Stable base:** wide footprint and casters with locks keep the holder stable near the bed/or table.
- **Accessory mounts / rings:** some stands offer adjustable rings or bowls to secure single basins or multiple receptacles; others provide splash guards or spigots to plumb waste to a container.

Incorporating Into Design

- **Design the basin to accept a quick-attach clamp:** integrate a keyed recess or bracket on the basin exterior that mates to a clamp on an or rail or a small adjustable arm. That allows the basin

to be moved to the limb rather than forcing the limb onto the basin.

- **Provide two modes:** (A) free-standing basin (standard) and (B) mountable basin with a universal clamp footprint compatible with common rail-mounts / small stands (use dimensions from Medline and similar stands for compatibility).
- **Include a locking tilt feature** (one-handed operation) to set pitch so fluid drains even when a limb rests on the cradle, design the lock to meet sterility and cleaning requirements.

Cons

- **Additional equipment:** stands/clamps increase or clutter, sterility challenges, and cost. Some facilities lack compatible rails or available footprint for stands.
- **Cleaning & sterilization:** clamps and articulated arms require design attention so they can be sterilized or covered with sterile drapes.
- **Compatibility:** universal mounting is hard; many use different rail systems, either adopt an adapter-based approach or offer multiple mounting options.

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